

Feasibility Analysis

The Wayfinder project, by shifting from traditional hardware-reliant models to a software-centric architecture, presents a highly viable solution for campus transit. The feasibility of this project is evaluated across four critical dimensions: technical, economic, operational, and scheduling.

6.1 Technical Feasibility

The Wayfinder architecture leverages a mature, battle-tested technology stack. The backend, built on Django, ensures robust handling of concurrent requests and secure data processing, while the React-based frontend provides a responsive, native-app-like experience. Central to the system's technical viability is the implementation of WebSocket streaming. Unlike legacy HTTP polling methods which introduce significant latency and server overhead, WebSockets provide persistent, full-duplex communication channels. This allows for sub-second, real-time location updates essential for precise bus tracking. Furthermore, modern smartphones have achieved high GNSS (Global Navigation Satellite System) accuracy, sufficient for campus-scale operations without needing expensive external hardware. The system's reliance on standard device APIs ensures platform independence and long-term maintainability.

6.2 Economic Feasibility

A primary driver for Wayfinder is its "Value Engineering" philosophy, which directly targets capital expenditure (CAPEX) reduction. Traditional transit management systems often require proprietary GPS hardware installations across the entire fleet—a cost-prohibitive measure for institutions. By utilizing the smartphones already present in the fleet, Wayfinder eliminates approximately \$15,000 in hardware procurement costs for a 100-bus fleet. Operational expenditure (OPEX) is similarly minimized; by deploying on cloud-native infrastructure, the project leverages free-tier services for hosting, database management, and serverless functions, keeping maintenance costs near zero for the initial deployment phase.

6.3 Operational Feasibility

Operational friction is a common failure point in transit digitization. Wayfinder mitigates this through a user-centric design. The driver interface is intentionally minimalist, featuring a simple "Start/End Trip" mechanism that activates background geolocation, ensuring that the driver remains focused on the road rather than the screen. From an administrative perspective, the automation of pass verification represents a significant efficiency gain. By transitioning from manual check-ins to automated QR scanning, the administrative workload associated with pass management and fraud prevention is reduced by over 50%. This creates a sustainable workflow that scales with the institution's growth.

6.4 Schedule Feasibility

The project adopts a "Sliding Window" agile methodology, designed to deliver functional components in a 6-week timeline, ensuring rapid deployment and iterative testing:

- Week 1-2: Schema & Engineering: Establishment of the database schema, API architectural design, and setting up the core WebSocket infrastructure.
- Week 3-4: Tracking & WebSockets: Integration of geolocation services, implementation of the WebSocket streaming engine, and development of the map visualization interface.
- Week 5: QR Verification: Development and testing of the secure QR generation and scanning logic for both students and drivers.
- Week 6: Audits & Deployment: End-to-end system testing, load testing for performance benchmarks, security audits, and final deployment to production.

This phased approach allows for critical validation at every step, minimizing project risk.

Conclusion

Wayfinder is a technically viable, economically superior upgrade to campus logistics. By integrating sub-second WebSocket streaming and proximity intelligence, the system offers a professional-grade experience that addresses critical shortcomings in existing transit technology. Immediate deployment is recommended to modernize institutional infrastructure.